

Course Recommendation System using NLP and TFIDF

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1 Introduction

The rapid growth of online learning platforms has created a need for personalized course recommendation systems. Given the diversity in learner backgrounds, especially professional learners, there's immense value in recommending courses tailored to an individual's experience, skills, and education. This project introduces a hybrid data-driven and NLP-based recommendation system that maps LinkedIn-style user profiles to a curated set of Coursera courses.

Two datasets are central to this system: a Coursera courses dataset[1] sourced from Kaggle containing course descriptions, skills, and metadata; and a LinkedIn profile dataset[2] with semi-structured fields like about, education, work experience, projects, and publications. We process and represent both as TFIDF vectors, enabling semantic comparison and personalized ranking of course options.

We use content-based filtering to identify relevant courses for a given user. This method is suitable due to the rich textual descriptions available in both datasets and the lack of user interaction history. Our pipeline includes pre-processing, feature extraction, and similarity computation.

2 Methodology

I began with data cleaning and normalization of both datasets. The Coursera dataset is preprocessed by removing the non-English courses, normalizing course descriptions and skill keywords, while the LinkedIn dataset required parsing nested JSON fields to extract structured features such as recent education degrees, last 3 work experience roles and descriptions, project descriptions and courses taken. The cleaned text was transformed via standard NLP preprocessing: converting to lowercase, removing punctuation and stopwords, and stemming tokens.

I used Document Term Matrix and TFIDF (Term Frequency-Inverse Document Frequency) vectorization to numerically represent both course descriptions and skills. Cosine similarity is applied on the reduced TFIDF matrix with PCA to get similar courses.

For LinkedIn users, I created synthetic documents by concatenating key profile elements (e.g., about me, education, projects, and experience summaries). Later, TFIDF Vectorizer was applied to both the course dataset and the LinkedIn synthetic data. Cosine similarity was computed between the LinkedIn profile vectors and the Coursera course vectors. The top-N courses with the highest similarity scores were suggested as personalized recommendations.

3 Results and Discussion

The system successfully generated relevant course recommendations for various user profiles. For example, users with a background in data science or engineering were often matched with courses in machine learning, Python programming, or cloud technologies. Users with business or planning roles saw recommendations aligned with project management and strategic decision-making courses.

EDA revealed key trends, supported by the following visual insights:

3.1 LinkedIn Profile Dataset

The graphs on the LinkedIn dataset show that nearly 70% of people are experienced and nearly 65% of people have shown their projects on LinkedIn. It emphasizes the growing importance of hands-on project experience in learning and showcasing skills.

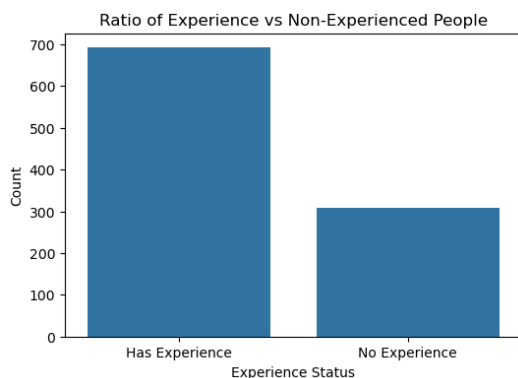


Figure 1: Ratio of Experience vs Non-Experienced People

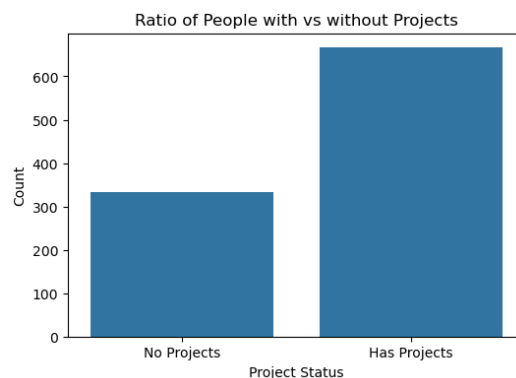


Figure 2: Ratio of People with vs without Projects

3.2 Coursera Dataset



Figure 3: Word Cloud of Skills



Figure 4: Word Cloud of Description

This word cloud visualizes the most common skills (left) and words (right), with larger text representing higher frequency. Prominent skills such as Python programming, Machine learning, Data analysis, and Data science stand out. Other notable skills include Communication, Cloud Computing, and Project Management, showing the value of both technical and soft skills. The word cloud (right) shows the key stem words in the course description with a frequency more than 70 with "data," "project," "apply," and "develop" standing out prominently.

3.3 Results of Recommending Similar Courses:

```
get_similar_courses('A life with ADHD', desc_similarity_matrix)
```

```

878             Social Psychology
859   School Health for Children and Adolescents
817             Psychological Research
959             Uncommon Sense Teaching
571   Introduction to Finance and Accounting
61             Animal Behaviour and Welfare
362             Foundations of Project Management
816             Psychological First Aid
943   The Science of Well-Being for Teens
759             Palliative Care Always
Name: course title, dtype: object

```

Figure 5: List of similar courses with TFIDF and PCA.

3.4 Results of Recommending Courses to LinkedIn profiles:

```
: rec = recommend_courses(user_id=100, top_n=5)
```

Recommendations for Omotolani K****u :

Recent Experience: Product Manager, Product Manager, Business Analyst
Recent Education: Philosophy

	course_title	similarity_score
265	Digital Transformation Using AI/ML with Google...	0.470805
796	Product Ideation, Design, and Management	0.427433
596	Introduction to Probability and Data with R	0.391563
883	Software Product Management	0.363806
264	Digital Technologies and the Future of Manufac...	0.332987

Figure 6: List of recommended courses with TFIDF model.

```
rec = recommend_courses_for_user_bert(100, similarity_matrix, top_n=5)
```

Recommendations for Omotolani K****u :

Recent Experience: Product Manager, Product Manager, Business Analyst
Recent Education: Philosophy

	course_title	similarity_score
796	Product Ideation, Design, and Management	0.631785
264	Digital Technologies and the Future of Manufac...	0.621508
967	User Experience Research and Design	0.574939
682	Marketing Mix Implementation	0.573899
883	Software Product Management	0.567607

Figure 7: List of recommended courses with BERT model.

The above results show the recommended courses for a LinkedIn profile. Figure 6 recommends courses based on the TFIDF model, and Figure 7 uses the BERT model as a baseline. Upon comparing both recommendations, TFIDF performs well and gives suitable courses but has less similarity scores compared to the BERT model.

4 Conclusion

Our recommendation engine demonstrates how textual data and NLP techniques, such as TFIDF and BERT embeddings, can be combined to create an effective personalized learning assistant. By analyzing user profiles and course content at the semantic level, the system can suggest relevant learning pathways tailored to an individual's professional background. Future improvements could enhance both precision and adaptability, for instance, incorporating collaborative filtering to leverage behavioral patterns or gathering user feedback to fine-tune recommendations. These additions would enable for a more dynamic, accurate and user-centric recommendation system.

References

- [1] Coursera Courses Dataset (Kaggle). <https://www.kaggle.com/datasets/siddharthm1698/coursera-course-dataset/data>.
- [2] LinkedIn Profile Dataset. <https://brihtdata.com/products/datasets>.